Lecture 16 – Most powerful tests for one-sided hypotheses

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Finishing example from Lecture 15:

Example 1 (Exponential rate)**.** Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathsf{Exp}(\theta)$, where $\theta > 0$ is the rate. Find the most powerful level α test for

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta = \theta_1,$$

where $\theta_1 > \theta_0$ and $0 < \alpha < 1$.

The likelihood function is

$$L_{\underline{x}}(\theta) = \theta^n \exp\left\{-\theta \sum_{i=1}^n x_i\right\} \mathbb{1}_{\{x_1>0,\ldots,x_n>0\}}.$$

So the likelihood ratio is

$$\frac{L_{\underline{x}}(\theta_1)}{L_{\underline{x}}(\theta_0)} = \left(\frac{\theta_1}{\theta_0}\right)^n \exp\left\{-(\theta_1 - \theta_0) \sum_{i=1}^n x_i\right\}.$$

Since $\theta_1 > \theta_0$, this ratio is decreasing in $\sum_{i=1}^n x_i$. By the Neyman-Pearson lemma, the most powerful test therefore rejects when $\sum_{i=1}^n X_i < c$, with c chosen to give size α. This makes sense because θ_1 being larger than θ_0 means the mean of $\mathsf{Exp}(\theta_1)$ is *smaller* than the mean of $\mathsf{Exp}(\theta_0)$.

Let us now choose c. Under H_0, we have that $\theta_0 X_i \overset{\text{iid}}{\sim} \mathsf{Exp}(1)$, so

$$\theta_0 \sum_{i=1}^n X_i \sim \mathsf{Gamma}(n, 1).$$

Let $F_{n,1}$ be the cdf of $\mathsf{Gamma}(n, 1)$ and let $q_\alpha = F_{n,1}^{-1}(\alpha)$ be its *lower-tail* quantile: $\mathbb{P}(Y < q_\alpha) = \alpha$ for $Y \sim \mathsf{Gamma}(n, 1)$. Then we should choose c so that

$$\alpha = \mathbb{P}_{\theta_0}\left(\sum_{i=1}^n X_i < c\right) = \mathbb{P}_{\theta_0}\left(\theta_0 \sum_{i=1}^n X_i < \theta_0 c\right) = \mathbb{P}(\mathsf{Gamma}(n, 1) \le \theta_0 c).$$

Thus we should set $\theta_0 c = q_\alpha$, or $c = q_\alpha/\theta_0$. We conclude that the most powerful level α test rejects when

$$\boxed{\sum_{i=1}^n X_i < \frac{q_\alpha}{\theta_0}.}$$

16.1 From most powerful to uniformly most powerful

The Neyman–Pearson lemma gives a most powerful (MP) test when both hypotheses are simple. For a composite alternative (in which Θ_1 consists of more than one parameter value), we want a single test that is most powerful at every alternative parameter value. The key question is therefore: *do the different simple alternatives all lead to the same test?*

Suppose first that we test

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta \in \Theta_1.$$

For each fixed $\theta_1 \in \Theta_1$, use Neyman–Pearson to find an MP level-α test against θ_1. Suppose the *same* test ψ^* works for every $\theta_1 \in \Theta_1$. We claim it is UMP level-α for the original hypotheses. To see why, fix any other level-α ψ for testing $H_0 : \theta = \theta_0$ vs $H_1 : \theta \in \Theta_1$, and fix any $\theta_1 \in \Theta_1$. Then ψ is also a valid level-α test for $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$. But ψ^* is MP for this test, so $\beta_{\psi^*}(\theta_1) \geq \beta_\psi(\theta_1)$. This is the definition of ψ^* being level-α for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \in \Theta_1$.

What if the null is composite too? As we now show, it suffices to solve the simple problems using one reference point θ_0 in the null, and then check that the resulting test has level α over the entire null.

Proposition 1 (From simple comparisons to a composite problem)**.** Consider $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta_1$. Suppose that there is some $\theta_0 \in \Theta_0$ and there is a test ψ^* that is MP of level α for $\theta = \theta_0$ versus $\theta = \theta_1$ for every $\theta_1 \in \Theta_1$. If also

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\psi^*(\underline{X}) = 1) \leq \alpha,$$

then ψ^* is UMP of level α for the composite problem.

Remark 1 (Which θ_0 should we try?)**.** Recall from the Neyman-Pearson lemma that the MP test ψ^* we know how to construct for simple hypotheses, θ_0 vs θ_1, has size exactly α. This means $\mathbb{P}_{\theta_0}(\psi^*(\underline{X}) = 1) = \alpha$ exactly. Since this same ψ^* is now required to satisfy $\mathbb{P}_\theta(\psi^*(\underline{X}) = 1) \leq \alpha$ for every θ in the null, the θ_0 in the null for which equality holds should be the one for which rejecting is most probable.

Proof. Let ψ be any competing level-α test for the composite problem. Because θ_0 belongs to the null, $\mathbb{P}_{\theta_0}(\psi(\underline{X}) = 1) \leq \alpha$. Thus ψ is also an allowable competitor in each simple problem. By the MP property,

$$\mathbb{P}_{\theta_1}(\psi^*(\underline{X}) = 1) \geq \mathbb{P}_{\theta_1}(\psi(\underline{X}) = 1) \qquad \text{for every } \theta_1 \in \Theta_1.$$

This is exactly the power comparison required for UMP. The extra size condition ensures that ψ^* itself is allowable for the composite problem. □

Remark 2. Enlarging the null makes the level constraint more demanding: a test must now control its rejection probability at more parameter values. It also leaves fewer competing tests. Thus an MP comparison proved using only θ_0 is enough, *provided our proposed test still satisfies the stronger level constraint.*

16.2 Monotone likelihood ratios

We now specify a property of the statistical model f_θ which guarantees that the same ψ_* is the level-α MP for θ_0 vs θ_1, regardless of the θ_1 in the alternative.

Definition 1. A family of pdfs or pmfs with scalar parameter θ has an **increasing monotone likelihood ratio (MLR)** in a statistic $T(\underline{x})$ if, for every $\theta_2 > \theta_1$,

$$\frac{L_{\underline{x}}(\theta_2)}{L_{\underline{x}}(\theta_1)}$$

depends on $\underline{x}$ only through $T(\underline{x})$ and is increasing (technically, non-decreasing) in $T(\underline{x})$. A **decreasing MLR** is defined in the same way, with increasing replaced by decreasing (technically, by non-increasing). When only the denominator is zero, interpret the ratio as $+\infty$. Sample points $\underline{x}$ such that $L_{\underline{x}}(\theta_1) = L_{\underline{x}}(\theta_2) = 0$ can be ignored.

An increasing MLR means that larger values of T provide stronger evidence for the larger parameter value. Crucially, *every pair* $\theta_2 > \theta_1$ orders the observations in the same way. This is what lets the simple Neyman–Pearson tests share a rejection rule.

Example 2 (Exponential families)**.** Suppose the likelihood has the form

$$L_{\underline{x}}(\theta) = h(\underline{x}) \exp\{b(\theta)T(\underline{x}) - c(\theta)\}$$

on a support that does not depend on θ. Then

$$\frac{L_{\underline{x}}(\theta_2)}{L_{\underline{x}}(\theta_1)} = \exp\{[b(\theta_2) - b(\theta_1)]T(\underline{x}) - [c(\theta_2) - c(\theta_1)]\}.$$

The last term is constant as the data vary. Thus the family has an increasing MLR if b is non-decreasing, and a decreasing MLR if b is non-increasing.

Example 3 (Exponential observations)**.** Suppose $X_i \overset{\text{iid}}{\sim} \mathsf{Exp}(\theta)$, where $\theta > 0$ is the *rate*. On the positive sample space,

$$L_{\underline{x}}(\theta) = \theta^n \exp\left\{-\theta \sum_{i=1}^{n} x_i\right\}.$$

For $\theta_2 > \theta_1$,

$$\frac{L_{\underline{x}}(\theta_2)}{L_{\underline{x}}(\theta_1)} = \left(\frac{\theta_2}{\theta_1}\right)^n \exp\left\{-(\theta_2 - \theta_1) \sum_{i=1}^{n} x_i\right\}.$$

This is decreasing in $T(\underline{x}) = \sum_{i=1}^{n} x_i$. A larger rate means shorter waiting times, so a small sum is evidence for a larger θ.

Example 4 (Uniform observations)**.** Suppose $X_i \overset{\text{iid}}{\sim} \mathsf{Unif}(0, \theta)$, where $\theta > 0$. On the positive sample space,

$$L_{\underline{x}}(\theta) = \theta^{-n} \mathbb{1}_{\{x_{(n)} < \theta\}}.$$

For $\theta_2 > \theta_1$, the ratio on the union of the two supports is

$$\frac{L_{\underline{x}}(\theta_2)}{L_{\underline{x}}(\theta_1)} = \begin{cases} (\theta_1/\theta_2)^n, & x_{(n)} < \theta_1, \\ +\infty, & \theta_1 \le x_{(n)} < \theta_2. \end{cases}$$

It is non-decreasing in $T(\underline{x}) = x_{(n)}$. Below θ_1, the ratio is constant. Once the maximum exceeds θ_1, the smaller parameter cannot explain the data at all, so we are completely sure the samples must have been drawn from $\mathsf{Unif}(0, \theta_2)$ rather than $\mathsf{Unif}(0, \theta_1)$.

16.3 UMP tests for one-sided hypotheses

Theorem 1 (Karlin–Rubin)**.** Consider

$$H_0 : \theta \le \theta_0 \qquad \text{versus} \qquad H_1 : \theta > \theta_0.$$

Suppose the family has an increasing MLR in T. If k is such that

$$\mathbb{P}_{\theta_0}(T > k) = \alpha,$$

then the test with rejection region $\mathcal{R} = \{\underline{x} : T(\underline{x}) > k\}$ is UMP of level α.

If instead the family has a decreasing MLR in T, the corresponding UMP test rejects when $T < k$, with $\mathbb{P}_{\theta_0}(T < k) = \alpha$.

Proof for increasing MLR.

Simple alternatives all give the same test: Fix any $\theta_1 > \theta_0$. By MLR, the likelihood ratio $L_{\underline{x}}(\theta_1)/L_{\underline{x}}(\theta_0)$ increases with T. Neyman–Pearson therefore tells us that the MP test for θ_0 vs θ_1 rejects when T is large. So the test is $\psi^*(\underline{x}) = \mathbb{1}_{\{T(\underline{x})>k\}}$, where k is chosen so that

$$\mathbb{P}_{\theta_0}(\psi^*(\underline{X}) = 1) = \mathbb{P}_{\theta_0}(T > k) = \alpha.$$

This value of k does not depend on θ_1! Therefore ψ^* is level-α MP for each $\theta_1 > \theta_0$.

Rejection probabilities increase as θ approaches θ_0 from below: We will show that

$$\mathbb{P}_\theta(T > k) \le \mathbb{P}_{\theta_0}(T > k) = \alpha \qquad \text{for every } \theta \le \theta_0. \tag{16.1}$$

Here, the equality to α is by design, and the inequality is what is new. We prove (16.1) below using the MLR property.

Conclude: We conclude that the test ψ^* is level α, since (16.1) gives

$$\sup_{\theta \le \theta_0} \mathbb{P}_\theta(T > k) = \mathbb{P}_{\theta_0}(T > k) = \alpha.$$

We have now checked both ingredients of Proposition 1: we found a test ψ^* which is MP for θ_0 against any $\theta_1 \in \Theta_1$, and we showed ψ^* has level α over the entire null. Hence it is UMP. For decreasing MLR, apply the same argument to $-T$. □

Proof of (16.1) *(optional).* First, we parse the likelihood ratio. The MLR assumption implies in particular that T is sufficient. Using the definition of sufficiency (recall Lec 3), we can write $f_\theta(\underline{x}) = h(\underline{x})g_\theta(T(\underline{x}))$, where g_θ is the pdf/pmf of T. Therefore, for any two parameter values θ_a, θ_b, we have

$$\frac{L_{\underline{x}}(\theta_b)}{L_{\underline{x}}(\theta_a)} = \frac{g_{\theta_b}(T(\underline{x}))h(\underline{x})}{g_{\theta_a}(T(\underline{x}))h(\underline{x})} = \frac{g_{\theta_b}(T(\underline{x}))}{g_{\theta_a}(T(\underline{x}))}.$$

Now assume $\theta_a < \theta_b$ and use g_a, g_b as shorthand for g_{θ_a}, g_{θ_b}. Increasing MLR gives that $t \mapsto g_b(t)/g_a(t)$ is nondecreasing in t. Therefore, if we take any two values $u < t$, we have

$$\frac{g_b(t)}{g_a(t)} \ge \frac{g_b(u)}{g_a(u)}, \qquad \text{so} \qquad g_b(t)g_a(u) \ge g_a(t)g_b(u).$$

We now take this inequality and integrate it over all $t > k$. This gives

$$\mathbb{P}_{\theta_b}(T > k)g_a(u) = \int_k^\infty g_b(t)g_a(u)dt \ge \int_k^\infty g_a(t)g_b(u)dt = \mathbb{P}_{\theta_a}(T > k)g_b(u).$$

Next, integrate over $u \le k$ to get

$$\begin{aligned}\mathbb{P}_{\theta_b}(T > k)\mathbb{P}_{\theta_a}(T \le k) &= \mathbb{P}_{\theta_b}(T > k) \int_{-\infty}^{k} g_a(u)du \\ &\ge \mathbb{P}_{\theta_a}(T > k) \int_{-\infty}^{k} g_b(u)du = \mathbb{P}_{\theta_a}(T > k)\mathbb{P}_{\theta_b}(T \le k)\end{aligned}$$

Write $p_a = \mathbb{P}_{\theta_a}(T > k)$ and $p_b = \mathbb{P}_{\theta_b}(T > k)$. The inequality is $p_b(1 - p_a) \ge p_a(1 - p_b)$; canceling the product terms gives $p_b \ge p_a$. Recalling what p_b and p_a are, we conclude that

$$\mathbb{P}_{\theta_b}(T > k) \ge \mathbb{P}_{\theta_a}(T > k) \qquad \text{for every } k, \qquad \text{for every } \theta_b > \theta_a.$$

Thus larger parameter values give larger upper-tail probabilities. This proves (16.1) □

Remark 3 (Other one-sided hypotheses)**.** The same test is also UMP when the null is just $\theta = \theta_0$. For the reversed hypotheses $H_0 : \theta \ge \theta_0$ versus $H_1 : \theta < \theta_0$, reverse the rejection direction: an increasing MLR gives a lower-tail test, and a decreasing MLR gives an upper-tail test. In each case, calibrate at θ_0.

Example 5 (Testing an exponential rate)**.** Suppose $X_i \overset{\text{iid}}{\sim} \mathsf{Exp}(\theta)$ and consider

$$H_0 : \theta \le \theta_0 \qquad \text{versus} \qquad H_1 : \theta > \theta_0.$$

The decreasing MLR in $T = \sum_{i=1}^{n} X_i$ gives the UMP rejection region

$$\mathcal{R} = \left\{ \underline{x} : \sum_{i=1}^{n} x_i < k \right\}, \qquad \mathbb{P}_{\theta_0}\left(\sum_{i=1}^{n} X_i < k \right) = \alpha.$$

This is the same test as for the simple null in the previous lecture. Smaller rates make waiting times longer, so rejecting for a small sum is less likely anywhere else in the null. Consequently, calibration at the boundary controls the whole null.

Example 6 (Testing a uniform endpoint)**.** Suppose $X_i \overset{\text{iid}}{\sim} \mathsf{Unif}(0, \theta)$ and consider

$$H_0 : \theta \ge \theta_0 \qquad \text{versus} \qquad H_1 : \theta < \theta_0.$$

The increasing MLR in $X_{(n)}$ tells us to reject for a small maximum. For $0 < k < \theta_0$, independence gives

$$\mathbb{P}_{\theta_0}(X_{(n)} < k) = \mathbb{P}_{\theta_0}(X_1 < k, \ldots, X_n < k) = \left(\frac{k}{\theta_0} \right)^n.$$

Equating this to α gives $k = \theta_0 \alpha^{1/n}$, so the UMP test rejects when

$$X_{(n)} < \theta_0 \alpha^{1/n}.$$

We can check the composite-null constraint directly: for every $\theta \ge \theta_0$,

$$\mathbb{P}_{\theta}(X_{(n)} < \theta_0 \alpha^{1/n}) = \left(\frac{\theta_0 \alpha^{1/n}}{\theta} \right)^n = \alpha \left(\frac{\theta_0}{\theta} \right)^n \le \alpha.$$

The same test is also UMP for the simple null $\theta = \theta_0$.

16.4 Why a two-sided alternative can prevent UMP

For a one-sided alternative, all simple alternatives favor the same tail. For a two-sided alternative, one side may favor unusually small observations and the other unusually large observations. A single test may then be unable to maximize power in both directions.

Proposition 2 (Uniqueness of the Neyman–Pearson test)**.** Suppose the two likelihoods $L_{\underline{x}}(\theta_0)$, $L_{\underline{x}}(\theta_1)$ are positive for the same $\underline{x}$'s, and the likelihood ratio has a continuous distribution under θ_0. If $k > 0$ is chosen so that the Neyman–Pearson test has size α, then this test is the **unique MP level-α test**.

The above conditions hold in the normal example below.

Example 7 (No UMP test for a two-sided normal mean)**.** Suppose $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, with $\sigma^2 > 0$ known, and $0 < \alpha < 1$. Consider

$$H_0 : \mu = \mu_0 \qquad \text{versus} \qquad H_1 : \mu \neq \mu_0. \tag{16.2}$$

If a UMP test exists, it must also be MP for θ_0 vs θ_1, for any $\theta_1 \neq \theta_0$. Fix $\theta_1 < \theta_0$. By the above proposition, the MP level-α test is unique, and given by the test ψ^* which rejects when $\bar{X} < \mu_0 - \frac{\sigma}{\sqrt{n}} z_\alpha$. where z_α is the upper-tail normal quantile, defined by $\mathbb{P}(Z > z_\alpha) = \alpha$ for $Z \sim \mathcal{N}(0, 1)$. Thus if the UMP exists, it must be this test. But consider its power at a $\mu_2 > \mu_0$. We have

$$\mathbb{P}_{\mu_2}(\psi^*(\underline{X}) = 1) = \mathbb{P}_{\mu_2}(\bar{X} < \mu_0 - \frac{\sigma}{\sqrt{n}} z_\alpha) < \mathbb{P}_{\mu_0}(\bar{X} < \mu_0 - \frac{\sigma}{\sqrt{n}} z_\alpha) = \alpha.$$

In contrast, the test ψ which rejects when $\bar{X} > \mu_0 + \frac{\sigma}{\sqrt{n}} z_\alpha$ is another level α test, and its power at μ_2 is

$$\mathbb{P}_{\mu_2}(\psi(\underline{X}) = 1) = \mathbb{P}_{\mu_2}(\bar{X} > \mu_0 + \frac{\sigma}{\sqrt{n}} z_\alpha) > \mathbb{P}_{\mu_0}(\bar{X} > \mu_0 + \frac{\sigma}{\sqrt{n}} z_\alpha) = \alpha.$$

Thus our only candidate ψ^* for being UMP is not actually UMP, since its power is smaller at μ_2 than another test's power. We conclude no UMP level-α test exists for (16.2).

Remark 4. Failure of UMP does not mean that there are no useful two-sided tests. It means that no single level-α test is most powerful at *every* alternative. The obstacle is the conflict between the best rejection rules for alternatives on opposite sides of the null.

To appreciate what's going on, consider the power functions in Figure 16.1, in which $\mu_0 = 0$. We consider three tests: one concentrates its power on the $\mu > 0$ region, rejecting if $\bar{X}_n$ is larger than $(\sigma/\sqrt{n})z_\alpha$. But then it naturally has low power in the $\mu < 0$ region, since large values of $\bar{X}_n$ are even less likely under $\mu < 0$ than under $\mu = 0$, for which we have *designed* the test to make this probability small! The second test concentrates its power on the $\mu < 0$ region, rejecting if $\bar{X}_n$ is sufficiently negative. Finally, the third test compromises between the two: it rejects when $|\bar{X}_n|$ is large. To make the Type 1 error small, we are forced to increase the cutoff on each side $|\bar{X}| > (\sigma/\sqrt{n})z_{\alpha/2} > (\sigma/\sqrt{n})z_\alpha$. Thus for positive $\bar{X}$'s, the rejection region is smaller than for the test that concentrated all its power on $\mu > 0$. Similarly for the test concentrating its power on $\mu < 0$.

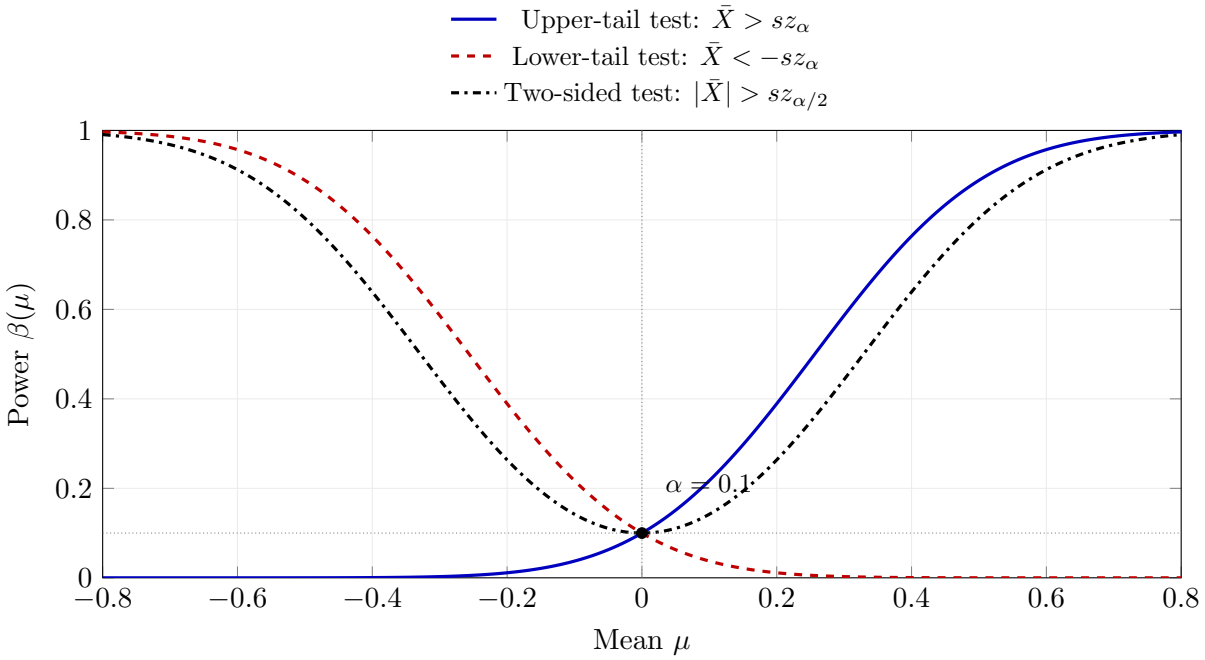


Figure 16.1: Normal power functions for $\mu_0 = 0$, $\alpha = 0.1$, $\sigma = 1$, and $n = 25$ (with $s = \sigma/\sqrt{n} = 0.2$). The horizontal axis is the mean μ. All three tests have rejection probability α at $\mu = 0$. Each one-sided test beats the two-sided test on its own side of the null.